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Mini-Project 3 — Full Interactive Power BI Dashboard

Country Development & Economic Performance Analysis

Grade: 100 marks

Dataset: Country Data Comparison.csv

Tool: Microsoft Power BI

Dataset Description:
The Country Data Comparison dataset contains socio-economic and development indicators for multiple countries. The dataset includes information related to:
• Population
• GDP
• Health indicators
• Life expectancy
• Inflation
• Imports and exports
• Birth and death rates
• Economic growth indicators

The dataset is highly suitable for:
• Interactive Power BI dashboards
• KPI reporting
• Country comparison analysis
• Trend analysis
• Geographical analysis
• Data storytelling
• Executive reporting

Recommended analytical dimensions:
• Country
• Region
• Economic Indicators
• Health Indicators
• Trade Indicators
• Population Statistics

Main Focus Areas

• Dashboard Design
• Interactive Reporting
• KPI Development
• Country Comparison Analysis
• Geographical Visualization
• Data Storytelling
• Drill-through Analysis
• Executive Reporting
• Power Query
• DAX Measures

Business Scenario

An international economic research organization wants to build a centralized analytical dashboard to monitor and compare development indicators across countries and regions.

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Senior management requires interactive reports to:
• Compare economic performance between countries.
• Monitor development indicators.
• Identify high-growth and low-growth regions.
• Analyze relationships between education, health, and economic factors.
• Support strategic policy and investment decisions.

Students are hired as junior business intelligence analysts to design a professional Power BI dashboard that helps decision-makers analyze country-level development data effectively.

Learning Objectives

By completing this project, students should be able to:
• Import and clean datasets using Power Query.
• Build interactive Power BI dashboards.
• Create KPI cards and DAX measures.
• Develop geographical and comparative visualizations.
• Apply data storytelling techniques.
• Build drill-through analytical pages.
• Generate analytical insights and recommendations.
• Design executive-level dashboards following visualization best practices.

Part 1 — Data Understanding & Preparation

Students must first inspect and understand the dataset structure.

Required tasks:
1. Explore dataset columns and understand indicator meanings.
2. Identify:
   • Numerical variables
   • Categorical variables
   • Missing values
   • Incorrect data types
   • Inconsistent country names

3. Import the dataset into Power BI.

4. Use Power Query to:
   • Remove duplicates
   • Handle missing values
   • Correct data types
   • Rename columns
   • Standardize country and region names
   • Remove unnecessary columns
   • Create conditional columns if needed

5. Create calculated columns:
   • GDP Category
   • Population Category
   • Development Level
   • Growth Indicator Category

Part 2 — Data Modeling

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- Development Level
- Growth Indicator Category

Part 2 — Data Modeling

Students must create a clean analytical model.

Required Tasks

1. Review the dataset structure and identify:
   - numerical fields,
   - categorical fields,
   - geographical fields,
   - development indicators.
2. Organize fields into logical categories:
   - Economic Indicators
   - Health Indicators
   - Population Indicators
   - Trade Indicators
3. Configure proper data types and formatting.
4. Create hierarchies such as:
   - Region → Country
   - Indicator Category → Indicator
5. Configure geographical fields correctly for map visualizations.
6. Create calculated groups/categories if needed:
   - GDP Level
   - Population Size
   - Development Classification

Expected Outcome

Students should produce a clean and optimized Power BI model suitable for:

- KPI reporting,
- country comparison analysis,
- interactive filtering,
- and dashboard storytelling.

Part 3 — KPI & DAX Development

Students must create KPI cards for:
• Average GDP
• Average Life Expectancy
• Total Population
• Highest GDP Country
• Lowest Inflation Country

Required DAX Measures:
• Average GDP
• GDP Growth Indicator
• Population Density Indicator
• Average Birth Rate
• Average Death Rate
• Trade Balance Indicator

Students should use:

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- Average Death Rate
- Trade Balance Indicator

Students should use:
- SUM
- AVERAGE
- CALCULATE
- DIVIDE
- IF
- FILTER
- RANKX

Students should also create:
- Dynamic dashboard titles
- Conditional formatting indicators
- Country ranking measures

Part 4 — Dashboard Development

Students must create a professional multi-page interactive dashboard.

Page 1 — Executive Overview

Required visualizations:
- KPI Cards
- GDP Comparison by Country
- Population Distribution
- Regional Development Comparison
- Country Ranking Table

Required filters:
- Region slicer
- Country slicer
- GDP category filter

Page 2 — Health & Education Analysis

Required visualizations:
- Life Expectancy Comparison
- Birth Rate vs Death Rate
- Health Indicator Analysis
- Scatter Plot for GDP vs Life Expectancy

Required features:
- Drill-through functionality
- Tooltips
- Conditional formatting

Page 3 — Economic Performance Analysis

Required visualizations:

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Required visualizations:
• Imports vs Exports
• Inflation Analysis
• Economic Growth Indicators
• Top and Bottom Performing Countries
• Regional Economic Comparison

Required features:
• Dynamic filtering
• Interactive navigation buttons
• Drill-through pages

Part 5 — Dashboard Design Requirements

Students must follow professional dashboard design principles.

Requirements:
• Consistent color theme
• Proper spacing and alignment
• Clear chart titles
• Readable labels and legends
• Proper visualization selection
• Logical dashboard navigation
• Clean executive-style layout

Students should avoid:

• Overcrowded dashboards
• Excessive colors
• Unnecessary text
• Misleading visualizations

Part 6 — Business Insights & Recommendations

Students must generate:
• 5 analytical insights
• 3 strategic recommendations

Examples of insights:
• Some regions show low population growth but high economic performance.
• Countries with lower inflation generally have higher life expectancy.

Examples of recommendations:
• Improve healthcare initiatives in countries with low life expectancy.
• Develop economic support strategies for high-inflation countries.

Deliverables

Students must submit:
1. Power BI File (.pbix)
2. Dashboard Screenshots
3. Presentation Slides
4. Business Insights Report
5. DAX Documentation
6. Data Cleaning Documentation

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Technical Requirements

Minimum requirements:
• 3 Dashboard Pages
• 6 Visualizations minimum
• 5 DAX Measures minimum
• 3 KPI Cards minimum
• 3 Interactive Filters
• 1 Drill-through Page
• 1 Map Visualization
• Dynamic titles
• Conditional formatting

Students must use:
• Power Query
• DAX
• Slicers
• Drill-through
• Tooltips
• Interactive visuals

Evaluation Rubric

Data Cleaning & Preparation: 20%
Data Modeling: 15%
DAX & KPI Development: 25%
Dashboard Design & Interactivity: 25%
Business Insights & Storytelling: 15%

Expected Skills After Completion

After completing this project, students should be able to:
• Build professional Power BI dashboards.
• Create executive KPI systems.
• Apply DAX calculations for analytical reporting.
• Design interactive visual reports.
• Perform country comparison analysis.
• Deliver business intelligence presentations professionally.

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